

PACKAGING, STORAGE, TRANSPORT, AND STERILIZATION EMBODIMENTS

In various embodiments, the intraluminal compliance and distance profiling device may be supplied with packaging and storage systems configured to preserve hygiene, maintain sensor integrity, and protect mechanical components during shipping, storage, and repeated consumer or clinical use.

The device may be packaged within sterile or semi-sterile enclosures including sealed polymer pouches, rigid protective trays, antimicrobial containers, or vacuum-sealed transport packaging. Packaging configurations may support single-use sterile deployment, reusable consumer storage, or research-grade distribution workflows.

In certain embodiments, storage systems may incorporate environmental protection features including moisture barriers, desiccant elements, UV-resistant materials, or antimicrobial coatings designed to preserve device cleanliness and prevent degradation of sensing surfaces.

Transport embodiments may include shock-absorbing structures, vibration isolation supports, or molded protective housings configured to prevent misalignment of sensitive sensing components or inflatable structures during shipping and handling.

The device ecosystem may further include smart storage enclosures capable of monitoring environmental conditions such as temperature, humidity, or contamination exposure. Storage systems may communicate with associated software to notify users of improper storage conditions or recommended maintenance actions.

In some implementations, sterilization-compatible materials may allow device components to undergo cleaning or sterilization processes including chemical disinfection, UV exposure, low-temperature gas sterilization, or other approved sanitation techniques compatible with embedded electronics and sensing elements.

Disposable protective barriers, sleeves, or tip covers may be supplied within packaging to support hygienic operation across multiple users or sessions. Packaging systems may organize consumables and replacement components for convenient access and controlled inventory management.

In certain embodiments, packaging may incorporate identification features such as QR codes, serialized labels, RFID tags, or cryptographic identifiers enabling traceability, authenticity verification, and supply-chain tracking.

The packaging and storage architecture may also serve as a docking or charging interface, allowing seamless transition between transport, storage, maintenance, and operational readiness without additional user handling steps.

Accordingly, packaging, storage, transport, and sterilization embodiments support safe distribution, reliable long-term use, and preservation of sensing accuracy while enhancing user convenience and maintaining hygienic standards across deployment environments.